Coordinated Population Forecast

2024

Through

2074

Areas Outside of Multnomah County UGBs

County UGB Coordinated Forecasts

June 25, 2024

Portland State
Population Research Center

PSUPRC

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1 Introduction

1.1 About this Report

This report was made in compliance with Oregon House Bill 2253 relating to population forecasts for land use planning. Portland State University Population Research Center is tasked with creating population projections for all Oregon counties except Multnomah, Clackamas, and Washington as well as population forecasts for all Urban Growth Boundaries (UGBs), except for the Metro area UGB. These population forecasts are generated using standardized demographic methods with input and feedback from county, city, and other local stakeholders.

1.2 Population Center Research Staff

- Neal Marquez, *Population Forecast Program Manager*
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1.3 How to Read this Report

This report should be read with reference to the documents listed below, which are downloadable on the Forecast Program website.

- Methods and Data for Developing Coordinated Population Forecasts: Provides a detailed description and discussion of the forecast methods employed. This document also describes the assumptions that feed into these methods and determine the forecast output.
- Forecast Tables: Provides complete tables of population forecast numbers by county and all sub-areas within each county for each five-year interval of the forecast period (2024-2074).

1.4 Suggested Citation

Marquez N., Sharygin E., Thangaraj S., Alkitkat H., Montcho G., Swanson D., Wilde J. (2024). Coordinated Population Forecast for Multnomah County, its Urban Growth Boundaries (UGB), and Area Outside UGBs 2024-2074. Population Research Center, Portland State University.

1.5 Acknowledgements

The PRC project staff wish to acknowledge and express gratitude for support from the Forecast Advisory Committee at the Oregon Department of Land Conservation and Development (DLCD) and the hard work of many people who contributed to the development of these forecasts by answering questions, lending insight, providing data, or giving feedback. The cover photo of Multnomah county was taken by Gary Halvorson as part of the Oregon State Photo Archives.

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2 Methodology

County populations were forecast using a variation of the cohort component method, wherein mortality, fertility, and migration were forecasted separately. Historic county populations were pulled from the National Cancer Institute Surveillance, Epidemiology, and End Results Program (SEER) U.S. County Population Data bridged race estimates 1990-2019. The SEER program provides annual estimates of every county in the United States broken down by age, sex, and broad racial and ethnic categories. These estimates are derived from population counts published by the U.S. Census Bureau with the added advantage that definitions of race and ethnicity are standardized through race bridging methodologies. Racial and ethnic categories include Hispanic, White, Black, Asian Pacific Islander, and American Indian Alaskan Native.

Death reports were collected from both the Oregon Health Authority (OHA) as well as the National Vital Statistics System (NVSS). All death records occurring among Oregon residents during the years 1990-2019 were tabulated by year, age, sex, county of residence, and racial or ethnic category matching the bridged race categories used in the SEER population data. Annual rates of mortality were calculated at the county level by age, sex, and bridged race category. Future rates of mortality were estimated out to the year 2074 using a modified version of the Lee-Carter Method developed by Li and Lee (2011). All counties in Oregon were modeled simultaneously such that state level trends partially informed future mortality rates for all counties and all bridged race groups.

Birth records were also obtained from both OHA and NVSS for the years 1990 to 2020. All births were tabulated by year, age of mother, county of residence of mother, and bridged race of mother. Annual rates of fertility were calculated at the county level by age and bridged race category of the mother. Similar to mortality, future rates of fertility were forecasted out to the year 2074 using the Li and Lee method, again modeling all counties simultaneously such that state level trends of fertility informed future county level estimates.

For both fertility and mortality forecasts, data pertaining to deaths and births occurring in the years 2020 through 2022 were not included in the model. This decision was made due to the impacts which the COVID-19 pandemic had on mortality and fertility rates during those years. We, and many other

demographers, anticipate the impacts of the COVID-19 pandemic on mortality and fertility to be shock effects from which regular rates will resume following the pandemic. Data pertaining to births and deaths in 2023 were not included as the data was not finalized at the time of this study.

Net migration counts for each county were estimated for 1990-2019 using the residual method. The residual method allows us to detail populations of net migrants by age, sex, and bridged race groups which are not published elsewhere. These data are then calibrated against official migration estimates published by the U.S. Census Bureau's American Community Survey Public Use Microdata Sample and Population Estimates Program. Future net migration estimates were made independently for each county in Oregon. Total net migration for each county was forecasted to 2074 using traditional time series methods. The demographic distribution of net migrants was taken as the average of the distribution observed over the 1990-2020 time period.

Fertility rates, mortality rates, and net migration counts were applied to the 2022 Oregon population, as estimated by PSUPRC, and aged forward such that projections were made up until the year 2074.

While county forecasts were generated for Multnomah county, the results are not included in this report. County and metro forecasts fall outside of the jurisdiction of the OPFP and PSUPRC. As such, only the areas outside if the Metro UGB are provided in this report. For Multnomah county and Metro population forecasts please see the Oregon Metro website here.

The PRC estimate formed the baseline of the forecast for individual UGBs, with the difference in population between incorporated city and UGB boundaries estimated based on assignment of population in individual census blocks in each county into a UGB area and or city area, or balance of county. Populations in individual UGBs or in the balance of county were forecast by projections of individual components of the housing unit method of population estimation. Historical rates of population and housing unit change since 1990 were used to generate a weighted average annual rate of change. Jurisdiction-level vacancy rates and average household size were held constant from the 2020 decennial census. Population forecasts for sub-areas were then controlled by the county-level forecasts, e.g., sub-area populations were allocated using the county total (top-down approach), and the population summation of the sub-areas does not exceed the county population.

Forecast Program surveys were used to make adjustments to the baseline results for counties and UGB areas. Recent development and plans obtained from surveys were generally implemented in the first 5-10 years of the forecast, except where they indicate a change in long-run outlook. For the immediate period (2023-2032), the development rate derived from the surveys or received reports was applied before 2030. If no planned housing units were reported, recent development rate (2010-2020) or the overall county rate was used. For the later period (2033-2074), housing unit growth was based on either a weighted average or an extrapolation of historic trend (1990-2020). Assumptions were made for individual cities based on knowledge obtained from the general surveys, housing surveys, as well as documentations (e.g., housing needs assessment, comprehensive development plans) received from the cities.

Many uncertainties still remain in understanding the climate change impacts on migration. Thus, specific scenarios of climate change, political unrest, or other shocks were not reflected in the current forecast. The forecast program methodology is described in further detail in an accompanying report available on the Population Research Center's website.

3 Historical Trends and Population Forecasts

3.1 Sub-Area Population

Sub-area populations within and outside the urban growth boundaries (UGBs) are forecasted using the housing unit method, and then adjusted to be consistent with the county level forecast. The areas outside of the Metro UGB have had a relatively stable population since 1990, however, population declined during the pandemic. In the forecast years we anticipate a continued decline of the population.

UGB	Estimate: 1990	Estimate: 2020	Estimate: 2022	Forecast: 2054	Forecast: 2074
Areas Outside UGB	10,085	10,534	8,663	7,698	6,958

Table 1: UGB population estimates and projections. Years prior to 2022 are population estimates while years after are population forecasts.

4 Glossary of Key Terms

- **Compound Annual Growth Rate (CAGR)**: The annualized rate of growth over a specific period of time. The CAGR is calculated using the formula $CAGR(t_0, t_n) = (\frac{P_t}{P_0})^{\frac{1}{t_n - t_0}} - 1$, where P_t is a value at time t, P_0 is a value at some initial time 0, and $P_t - P_0$ is the difference in years between two time periods.
- **Race Bridging**: Making data collected using one set of race categories consistent with data collected using a different set of race categories, to permit estimation and comparison of race-specific statistics at a point in time or over time.
- **Cohort-Component Method**: A method used to forecast future populations based on a baseline or starting population, and cumulative changes in births, deaths, and migration.
- **Coordinated population forecast**: A population forecast prepared for the county and sub-county jurisdictions including urban growth boundary (UGB) areas and all non-UGB area in the balance of county.
- **Group quarters**: The US Census Bureau defines group quarters as places where "people live or stay in a group living arrangement that is owned or managed by an organization providing housing and/or services for the residents". Examples of a group quarter may include college dorms, skilled nursing facilities, groups homes, prison, etc.
- **Housing unit**: A house, apartment, mobile home or trailer, group of rooms, or single room that is occupied or is intended for occupancy.
- **Housing-Unit Method**: A method used to estimate current populations or forecast future populations based on changes in housing units, vacancy rates, the average numbers of persons per household (PPH), and group quarters population counts.
- **Persons per household (PPH)**: The average household size (i.e., the average number of persons per occupied housing unit).
- **Total Fertility Rate (TFR)**: The number of children a woman would have by the end of a defined childbearing age. In this report, child-bearing age is from 15 to 49.
- **Life Expectancy**: The average number of years of life expected at birth based on the current set of age-specific mortality rates.

5 Appendix A: General Survey for Oregon Forecast Program

Respondent	Question	Response
None	Username	None
None	Jurisdiction	None
None	1. Observations about Housing (Vacancy rates, seasonal occupancy, demolitions, renovations)	None
None	2. Planned Housing Developments or Group Quarters Facilities.	None
None	3. Observations about Population (e.g. birth rates, aging, immigration, racial and ethnic change)	None
None	4. Economic Development (e.g. new employers or facilities, including number of jobs and est. year of completion)	None
None	5. Infrastructure Projects (e.g. transportation and utilities)	None
None	6. Other Factors Promoting Population or Housing Growth	None
None	7. Other Factors Hindering Population or Housing Growth	None
None	8a. Summary of current or proposed policies affection growth in your jurisdiction.	None
None	8b. Findings related to growth or population change from studies conducted in you jurisdiction.	None
None	8c. The effects of wildfires or other disasters in your jurisdiction on housing, employment/economics, and infrastructure.	None
None	8d. The effects of the COVID-19 pandemic and policy measure on employment and current and planned developments.	None
None	9. For representatives from counties only: we invite you to provide tax lot data if available. These may be sent via email to askprc@pdx.edu OR if your data is available on-line please enter the URL below.	None
None	Comments?	None

Table 2: Questionaire Response 1